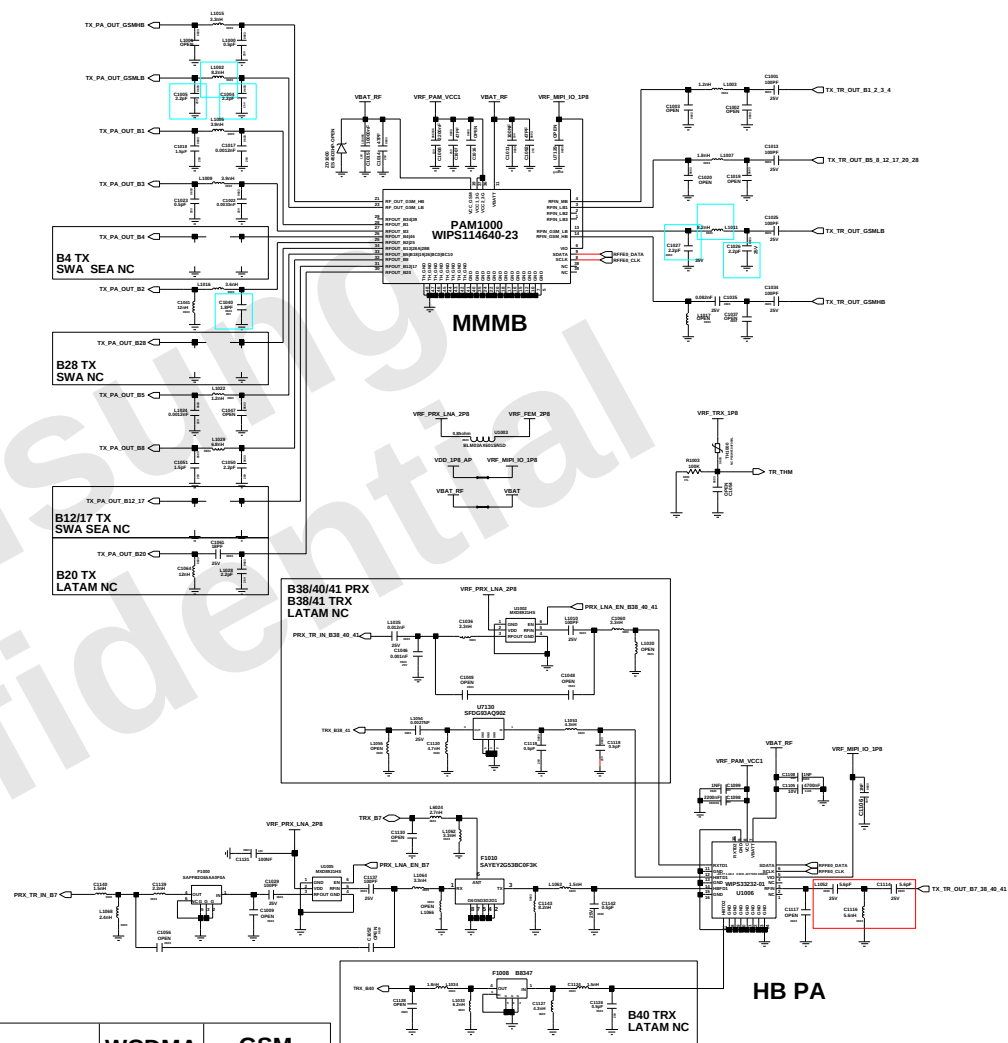
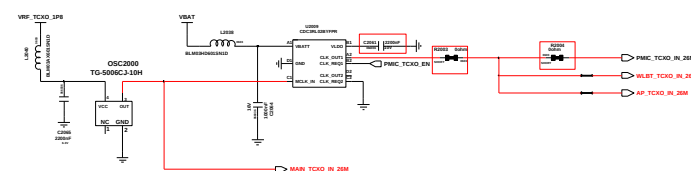


	LTE	WCDMA	GSM
SWA / CIS / MENA / AFR	B1/2/3/5/7/8/20/38/40/41	B1/2/5/8	QUAD
SEA	B1/3/5/7/8/20/28/38/40/41	B1/2/5/8	QUAD
LATAM	B1/2/3/4/5/7/8/12/17/28	B1/2/4/5/8	QUAD
KOR	B1/3/4/5/7/8/17/20/28/38/40/41	B1/2/4/5	QUAD



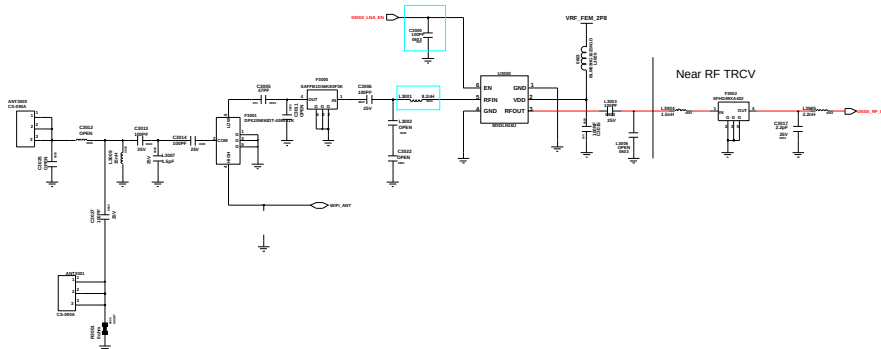
Engineer	MAIN RF TITLE: SM-A105F	
Drawn by:		
RED CHK:		
ISO CTR CHK:		
INFO ENGR CHK:		
QA CHK:	Drawing Number	

DRX

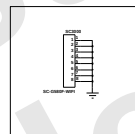
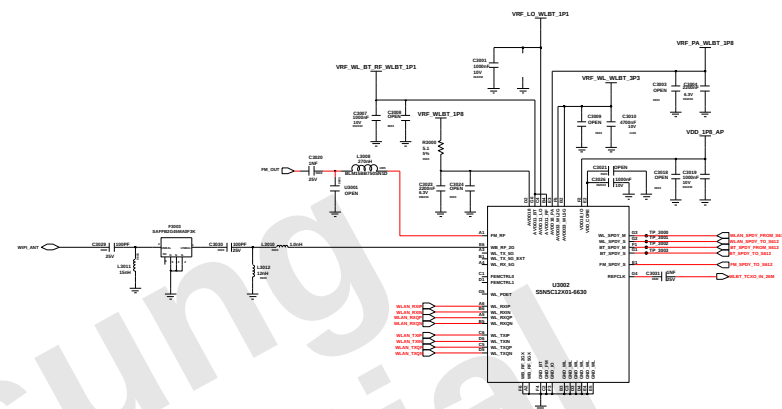


user swcad71 10.252.50.122 20190222-135049
--

GNSS



BT/WIFI



Engineer:	WIFI/BT/GPS		
Drawn by:			
RED CHK:	TITLE:		
DOC & TRS CHK:	SM-A105F		
MFG ENGRG CHK:			
QA CHK:	REV	Drawing Number	Job

POP LP4 2G : SEC 1105-002944
POP LP4 3G : SEC 1105-002962

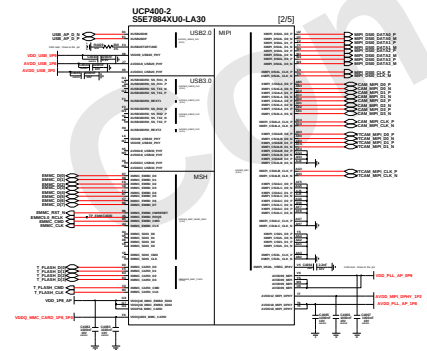
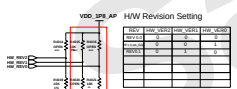
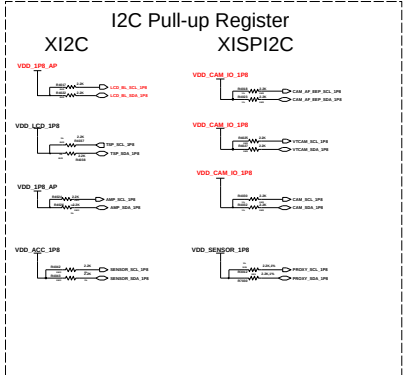
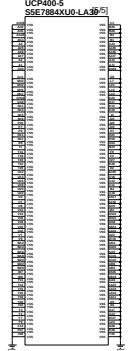
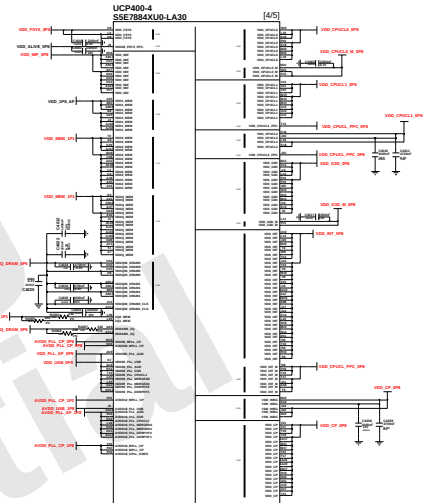
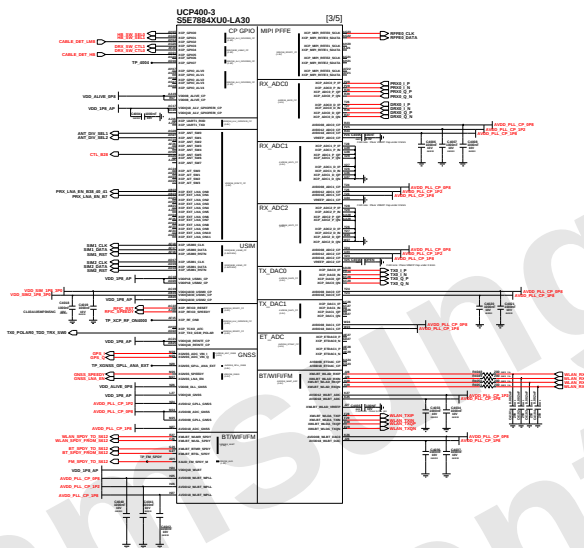
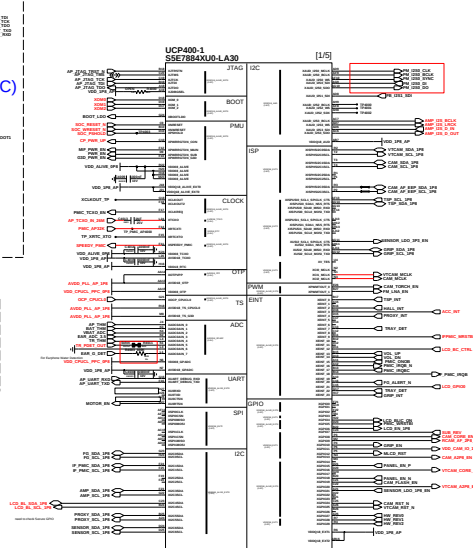
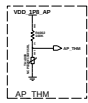


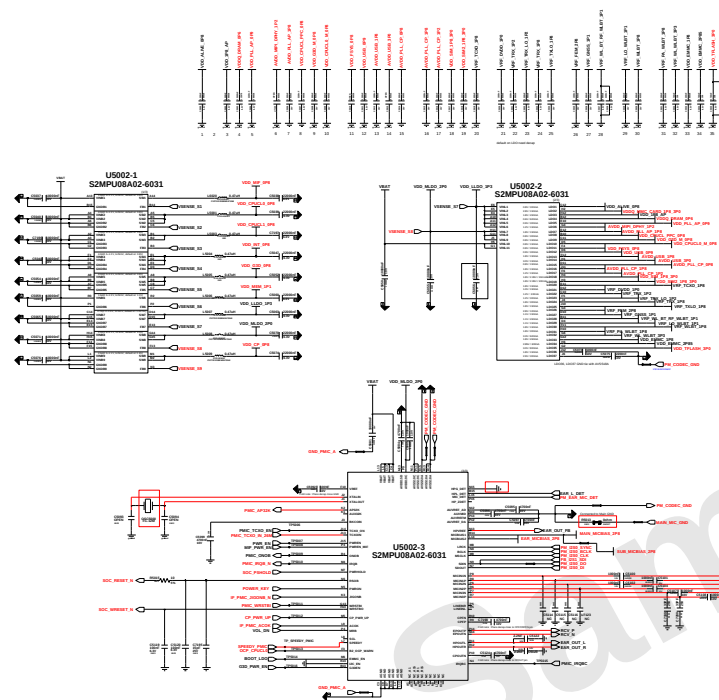
Figure 10 illustrates recommended decoupling capacitor placement for various power planes and components. The diagrams are organized into three rows and four columns:

- Row 1:**
 - VDD_CPUCL0 SPI:** Shows a capacitor C1 connected between VDD_CPUCL0 and GND.
 - VDD_CPUCL1 SPI:** Shows a capacitor C1 connected between VDD_CPUCL1 and GND.
 - VDD_MIF SPI:** Shows a capacitor C1 connected between VDD_MIF and GND.
 - VDD_INT SPI:** Shows a capacitor C1 connected between VDD_INT and GND.
 - VDD_CPUCL_PPC SPI:** Shows a capacitor C1 connected between VDD_CPUCL_PPC and GND.
- Row 2:**
 - VDD_GSD SPI:** Shows a capacitor C1 connected between VDD_GSD and GND.
 - VDD_CP SPI:** Shows a capacitor C1 connected between VDD_CP and GND.
 - VDD_1PE AP:** Shows a capacitor C1 connected between VDD_1PE and GND.
 - VDD_MEM SPI1:** Shows a capacitor C1 connected between VDD_MEM and GND.
- Row 3:**
 - VDD_PLL AP SPI:** Shows a capacitor C1 connected between VDD_PLL and GND.
 - AVDD_PLL AP SPI:** Shows a capacitor C1 connected between AVDD_PLL and GND.
 - AVDD_PLL CP SPI:** Shows a capacitor C1 connected between AVDD_PLL and GND.
 - AVDD_PLL CP SPI:** Shows a capacitor C1 connected between AVDD_PLL and GND.

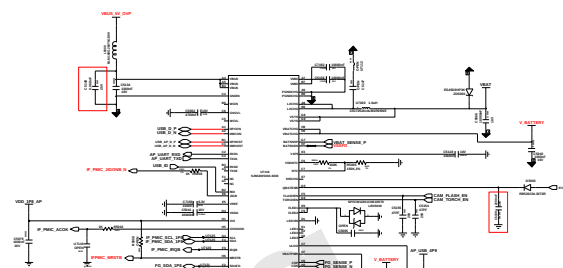
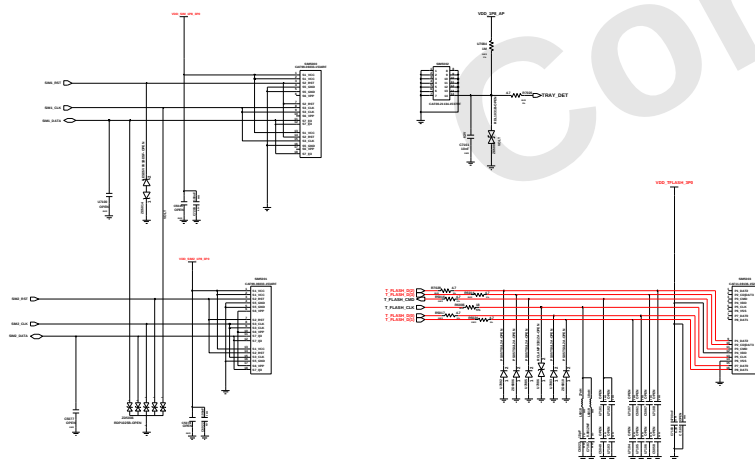
Each diagram includes labels for the power plane, the capacitor, and the connection points (e.g., BALL, BALL1, BALL2).

POWER

PMIC



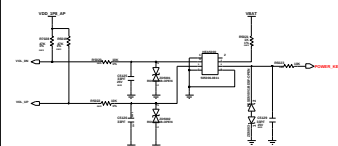
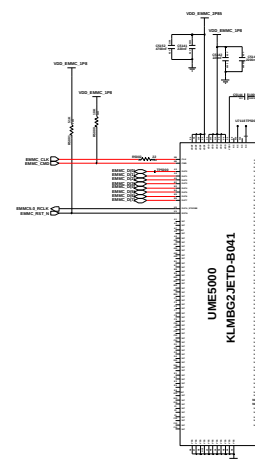
SIM & SD CARD (3IN3)



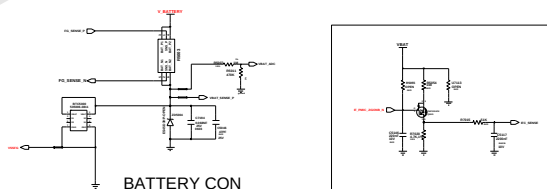
IF PMIC EVT4(CHG+FG+BUCK)



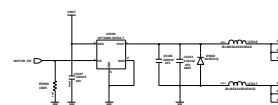
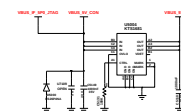
16G : SEC (1107-002548)
32G : SEC (1107-002552)



VOLUME KEY POWER KEY



BATTERY CON

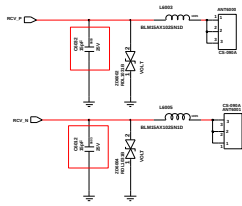


MOTOR

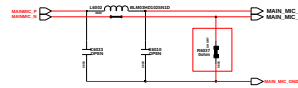
Project	PMIC, 4MMC
Manager	
Version	
Release Date	
Next Review Date	SM-A105F

SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPERTY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO., LTD. PROPERTY
AND NOT BE LOANED TO OR REPRODUCED FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

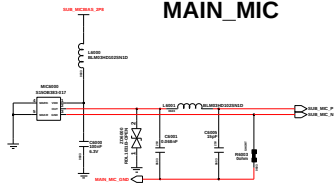
AUDIO



RCV Contact

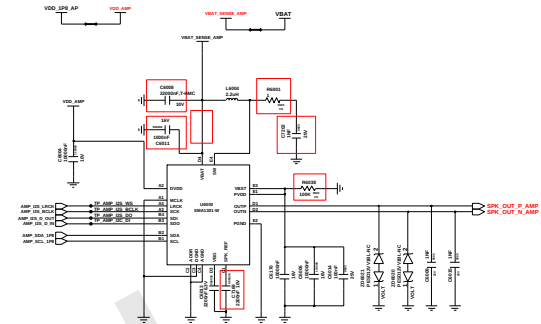
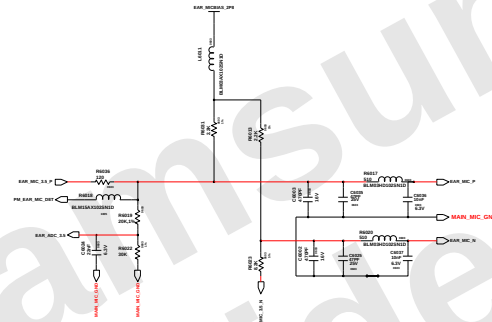


MAIN_MIC

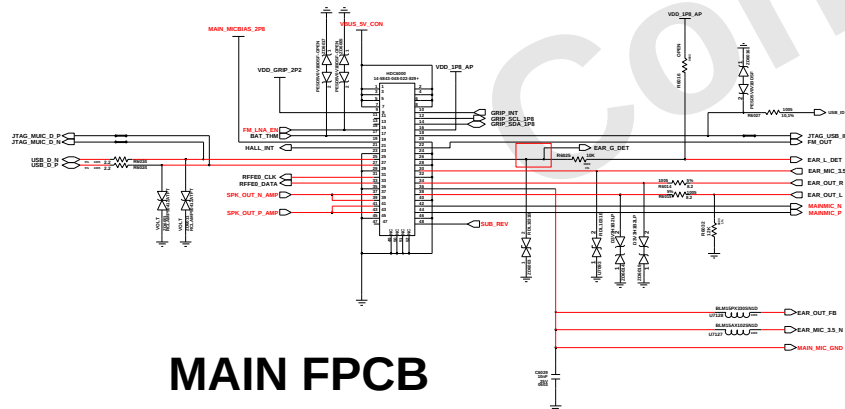


Sub MIC

EAR MIC



Audio amp



MAIN FPCB



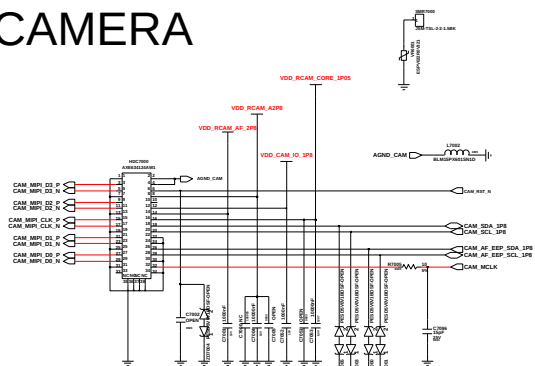
Grip Sensor_LDO

Document	AUDIO	
Version	1.0	1
File Name	SM-A105F	
File Path	SM-A105F	
File Size	1.0	
File Date	20190222-135053	

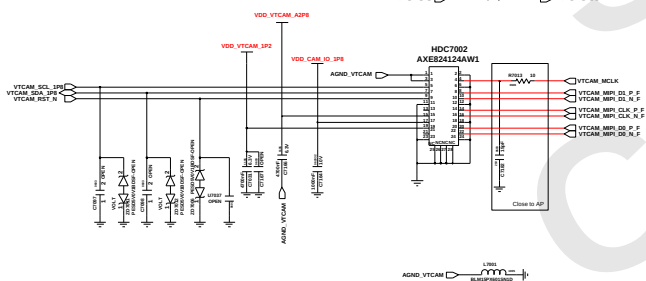
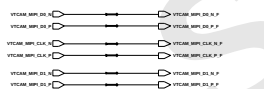
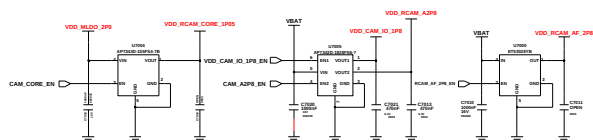
SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO.'S PROPERTY
IT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

CAMERA

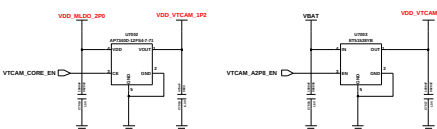
CAM DECO GASKET



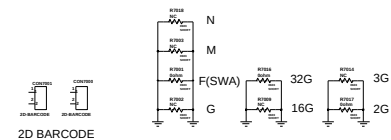
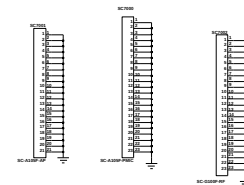
MAIN_CAMERA (13M) : SM-M205F



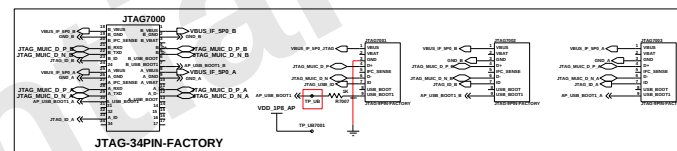
FRONT_CAMERA (5M) : SM-M105F



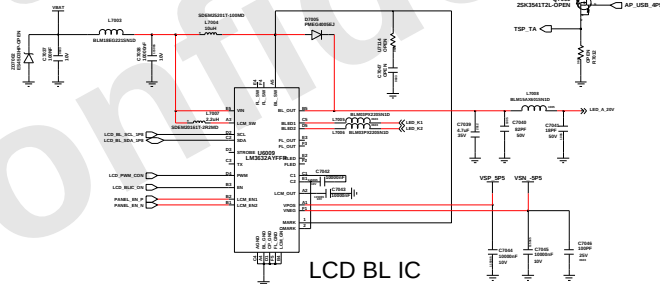
SHIELD CAN



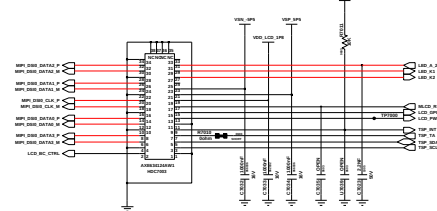
PCB JTAG



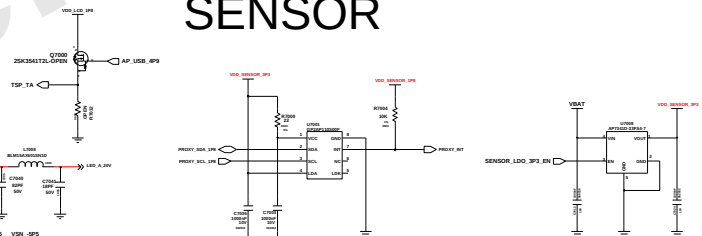
DISPLAY (6.22" INCELL LCD)



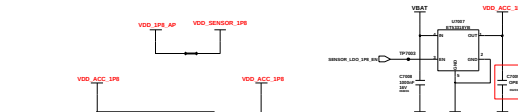
6.22" INCELL TFT LCD



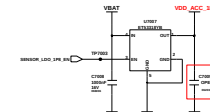
SENSOR



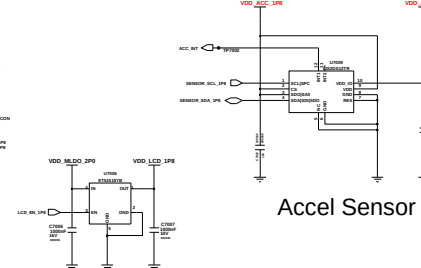
Proxy Sensor



SENSOR LDO



Accel Sensor



Engineer	CAMERA, DISPLAY, SENSOR		
Drawn by			
Read Ck:	TITLE:		
DWG. LTRG. Ck:	SM-A105F		
MFG. INSTR. Ck:			
Original	CAT. Ck:	REV:	Drawing Number

SM-A105F

U7001

BH7002

BH7005

HDC7002

SIM5002

SIM5003

SIM5000

SIM5001

SC7001

HDC7000

UCP400

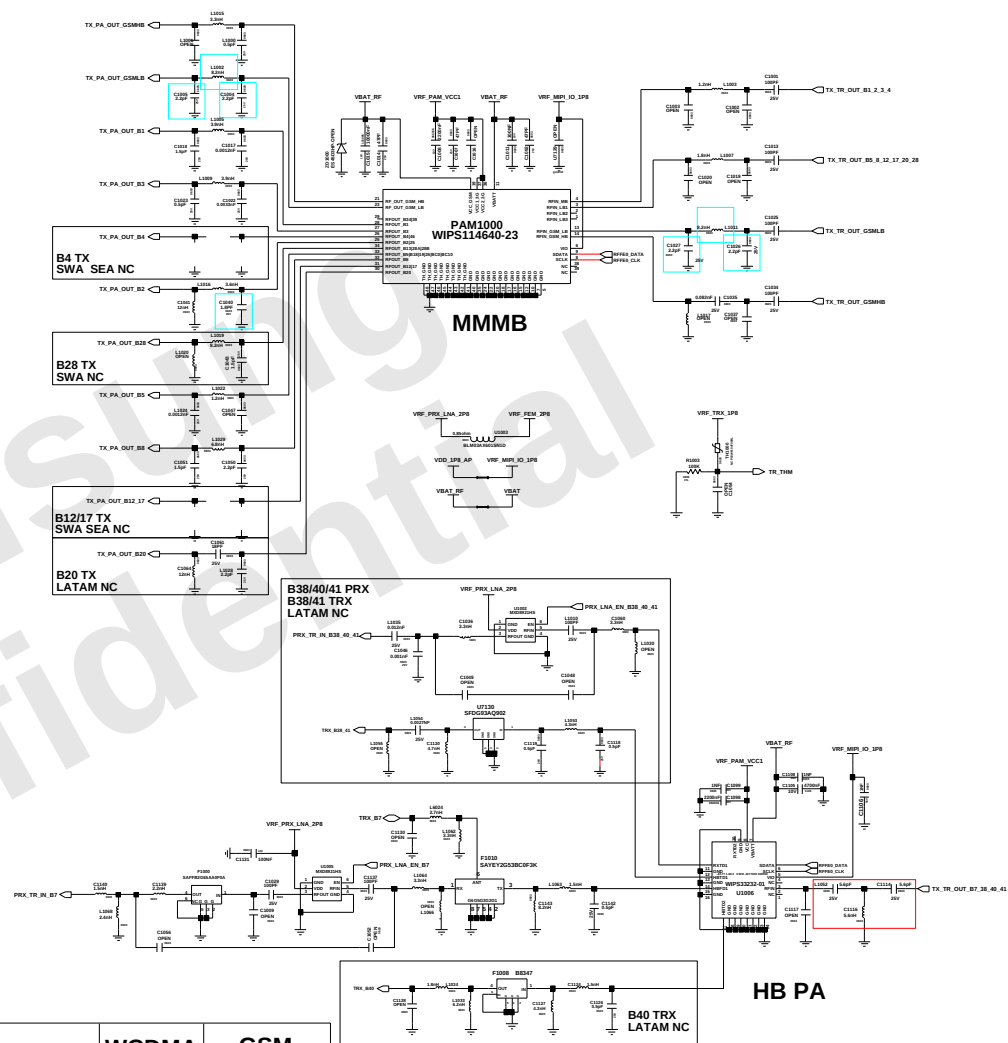
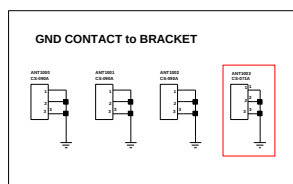
HDC4000

CON7000

BH7001

BH7004

JTAG7001

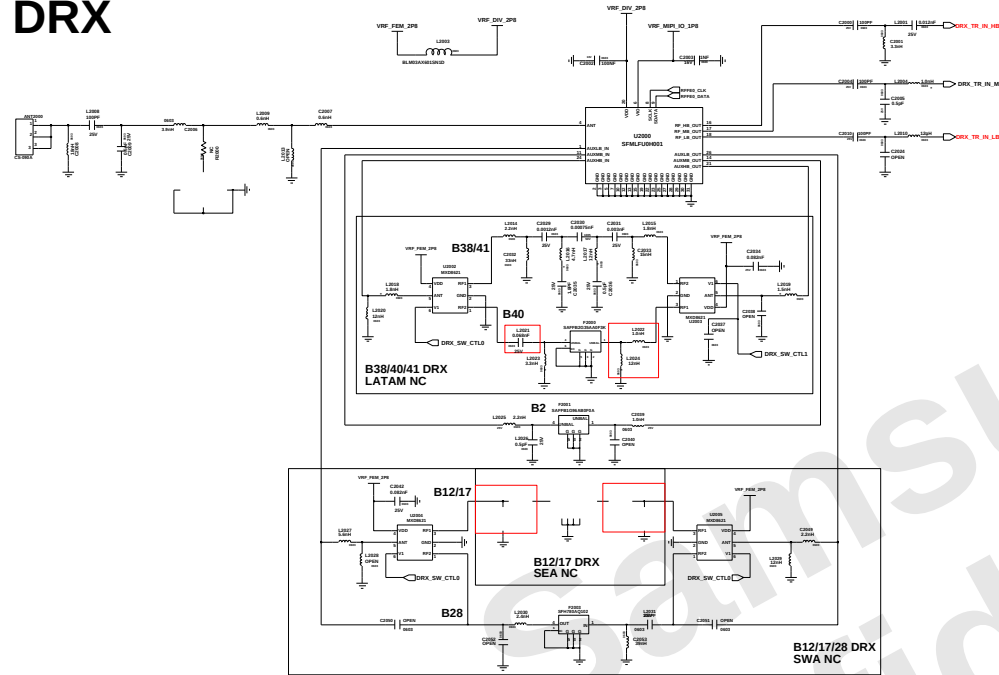


	LTE	WCDMA	GSM
SWA / CIS / MENA / AFR	B1/2/3/5/7/8/20/38/40/41	B1/2/5/8	QUAD
SEA	B1/3/5/7/8/20/28/38/40/41	B1/2/5/8	QUAD
LATAM	B1/2/3/4/5/7/8/12/17/28	B1/2/4/5/8	QUAD
KOR	B1/3/4/5/7/8/17/20/28/38/40/41	B1/2/4/5	QUAD

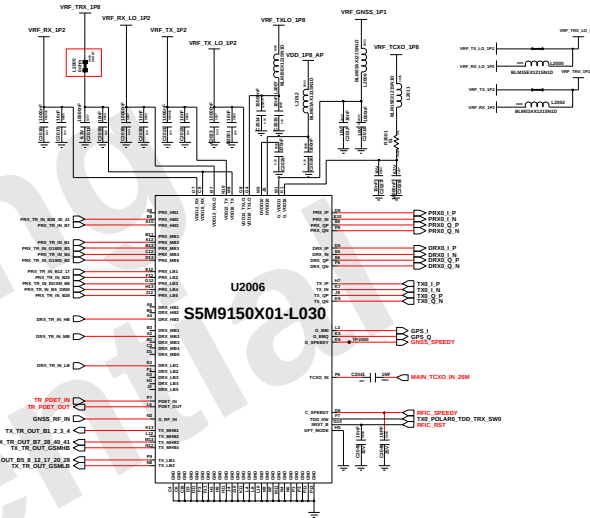
Engineer	MAIN RF			
Drawn by:				
RED CDR:				TITLE:
DOC CTRG. CDR:				SM-A105F
MPD KTRNG CDR:				
Langsd	QA CDR:	Revision Number	400	

SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPERTY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO.'S PROPERTY
AND NOT BE LOANED TO OR REPRODUCED FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

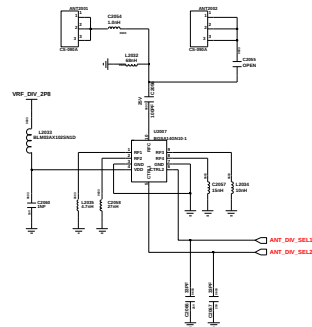
DRX



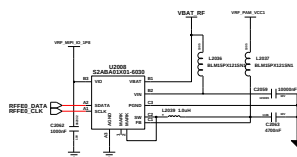
TRANSCIEVER nCA



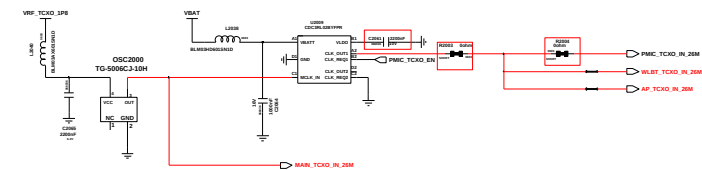
DRX GND SWITCH



APT MODULATOR

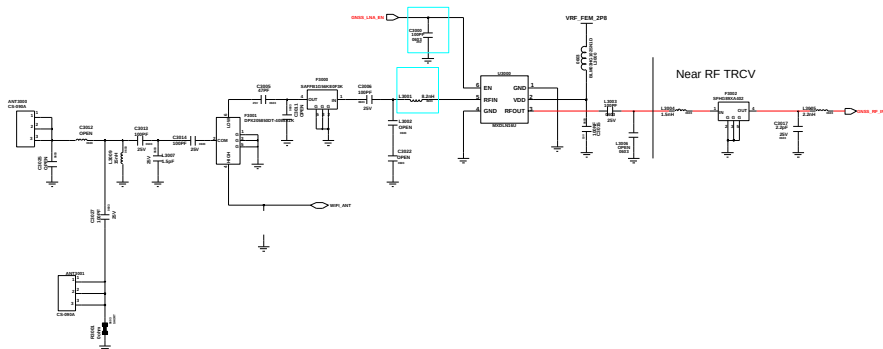


TCXO

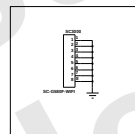
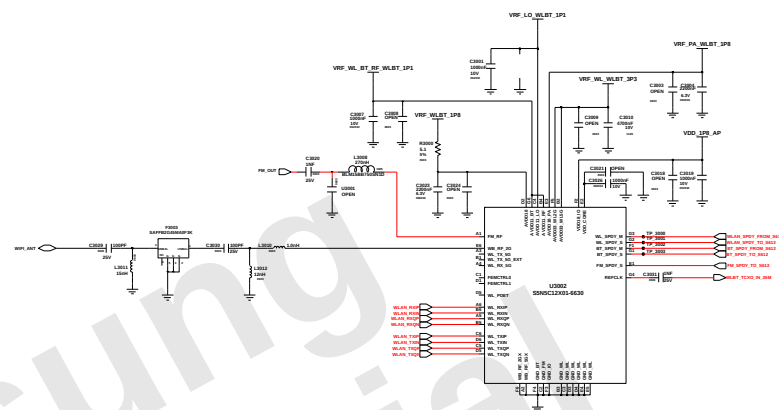


Engineer	DRX, TRANSCIEVER
Designer	
Checker	
Reviewer	
Approval	
Signature	
Date	

GNSS

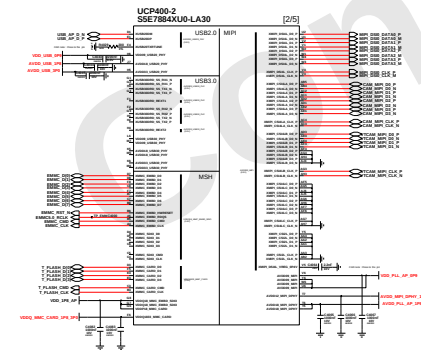
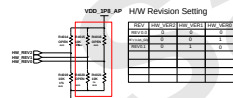
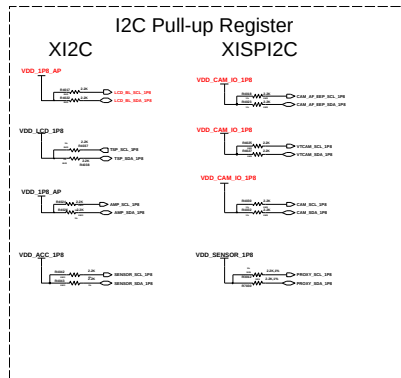
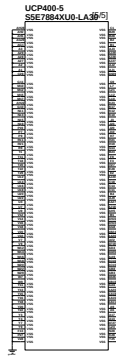
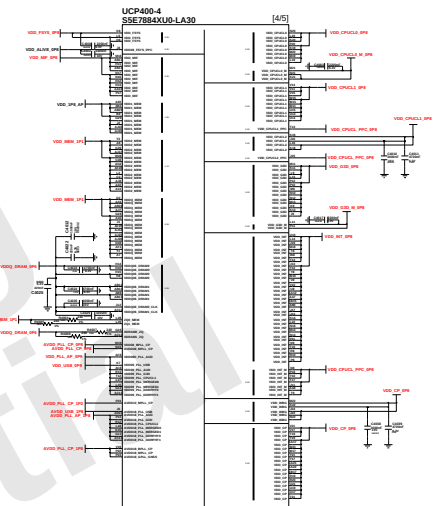
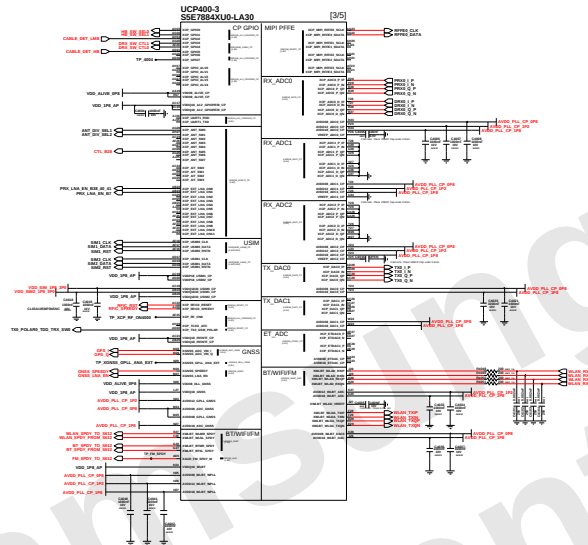
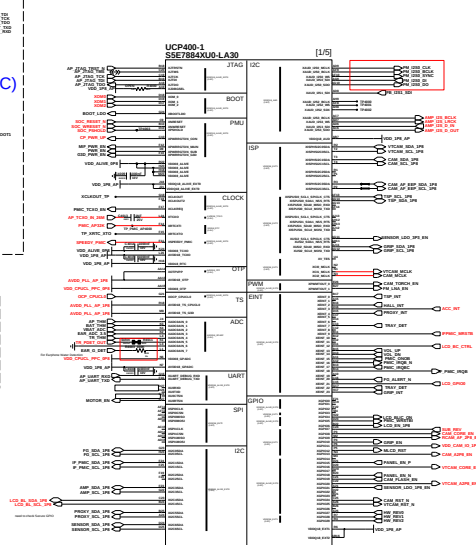
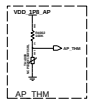


BT/WIFI



Engineer	WIFI/BT/GPS
Drawn by	
RED CHK	
DOC CTRL CHK	
APPROV ENG CHG	TITLE: SM-A105F

POP LP4 2G : SEC 1105-002944
POP LP4 3G : SEC 1105-002962



VDD_CPUCLK1_SPS

VDD_SEF_SPS

VDD_INT_SPS

VDD_CPUCLK_PPC_SPS

VDD_GSD_SPS

VDD_CP_SPS

VDD_1PS AP

VDD_MEM_SPS1

VDD_MEM_SPS1

VDD_PLL_AP_SPS

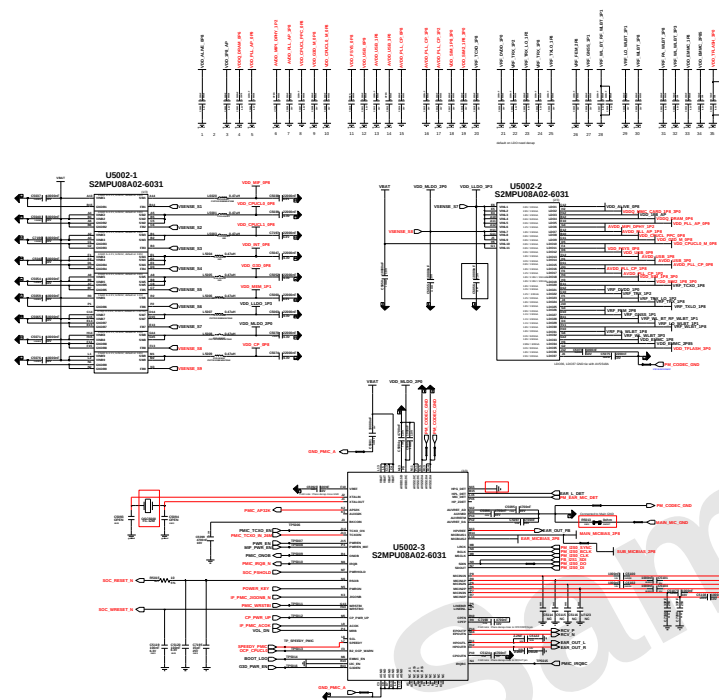
VDD_PLL_CP_SPS

VDD_PLL_CP_SPS

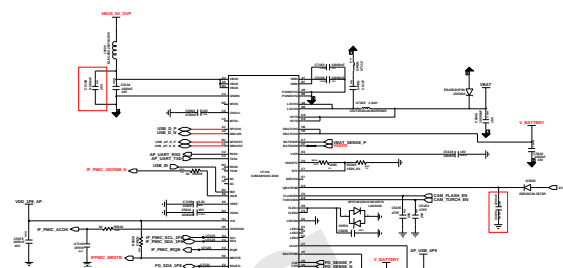
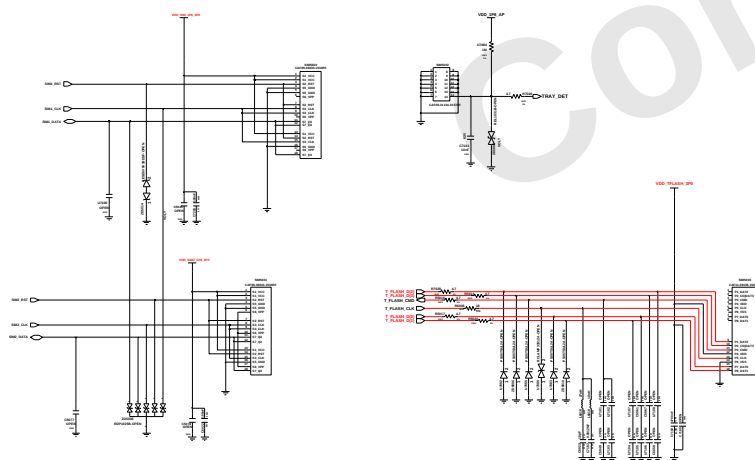
VDD_PLL_CP_SPS

POWER

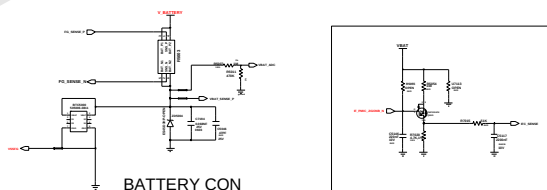
PMIC



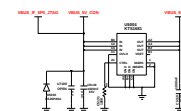
SIM & SD CARD (3IN3)



IF PMIC EVT4(CHG+FG+BUCK)

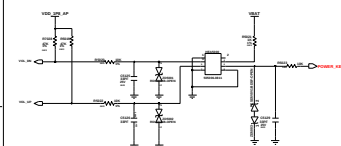
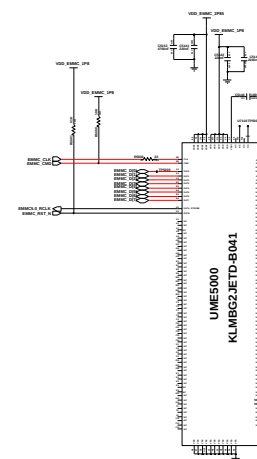


BATTERY CON

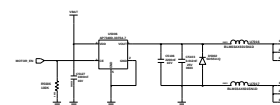


eMMC

16G : SEC (1107-002548)
32G : SEC (1107-002552)



VOLUME KEY POWER KEY

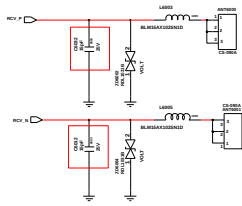


MOTOR

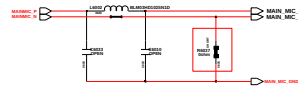
Project:	PMC, eMMC
Model:	
Part:	
Rev:	
Doc: 1000-0000	SM-A105F

SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPERTY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO., LTD. PROPERTY
AND NOT BE LOANED TO OR REPRODUCED FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

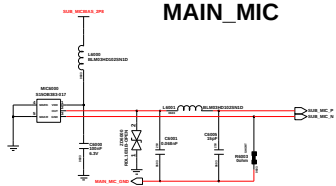
AUDIO



RCV Contact

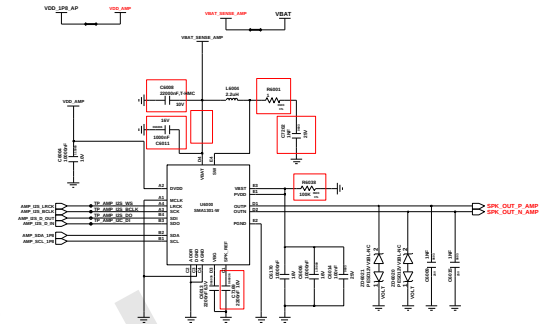
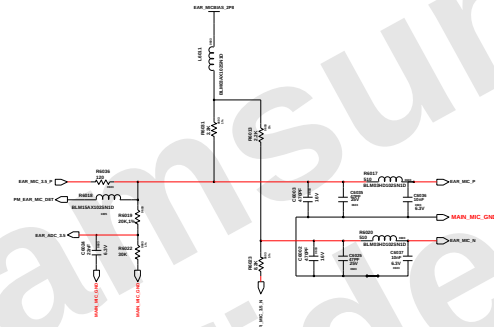


MAIN_MIC

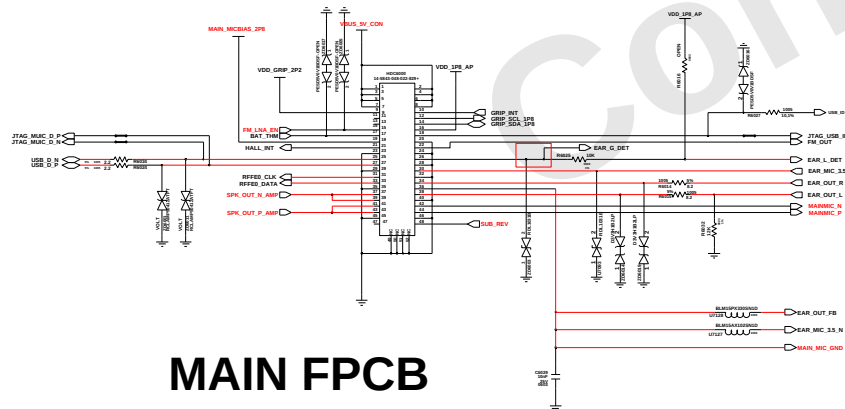


Sub MIC

EAR MIC



Audio amp



MAIN FPCB



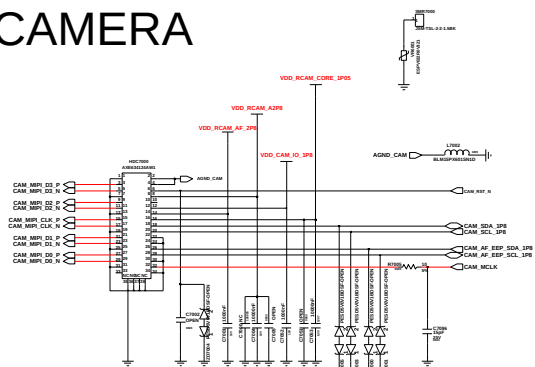
Grip Sensor_LDO

Document	AUDIO	Page	1
Document No.	SM-A105F	Page	1
Document Title	SM-A105F	Page	1
Document Date	20190222-133936	Page	1

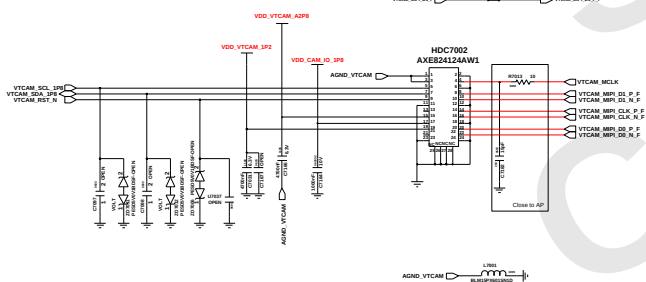
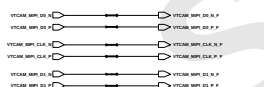
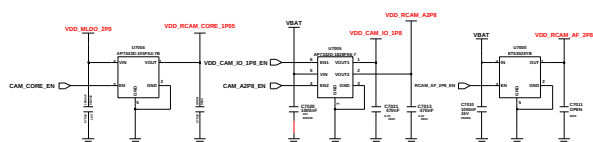
SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO.'S PROPERTY
IT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

CAMERA

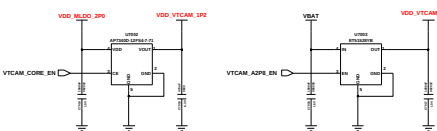
CAM DECO GASKET



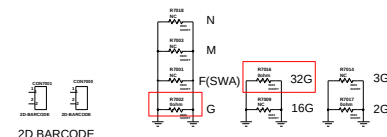
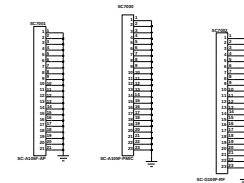
MAIN_CAMERA (13M) : SM-M205F



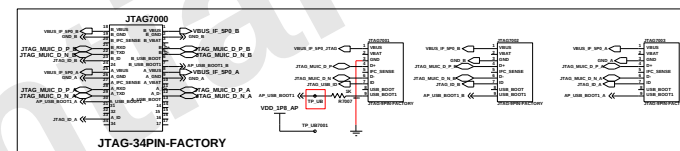
FRONT_CAMERA (5M) : SM-M105F



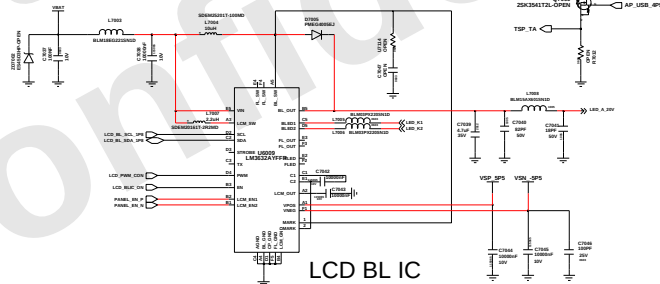
SHIELD CAN



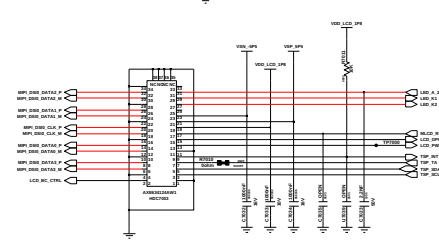
PCB JTAG



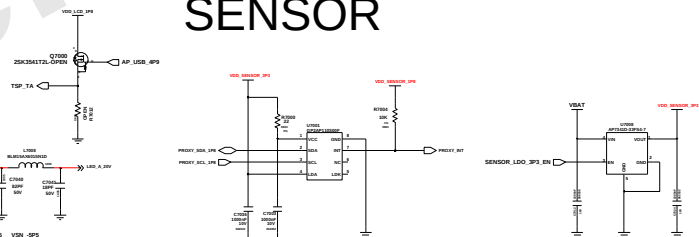
DISPLAY (6.22" INCELL LCD)



6.22" INCELL TFT LCD



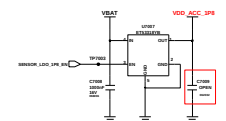
SENSOR



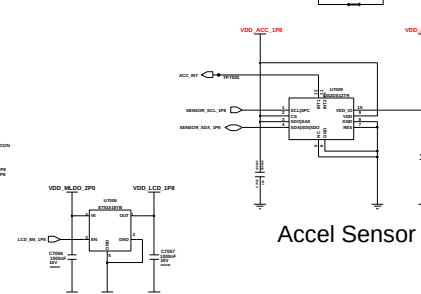
Proxy Sensor



SENSOR LDO



Accel Sensor



Designed by	CAMERA, DISPLAY, SENSOR		
Drawn by			
Field CDR:	TITLE:		
DWG. CTR. CHK:	SM-A105F		
MFG ENGR. CHK:			
Original	QA CDR:	REV:	Drawing Number

SM-A105G

U7001

BH7002

BH7005

HDC7002

SIM5002

SIM5003

SIM5000

SIM5001

SC7001

HDC7000

UCP400

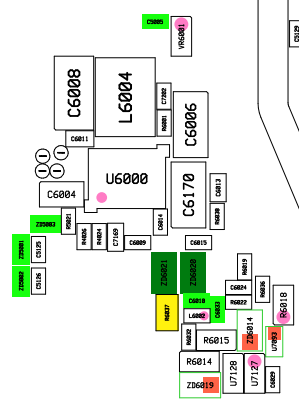
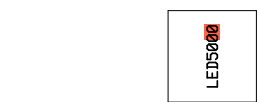
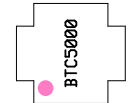
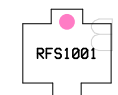
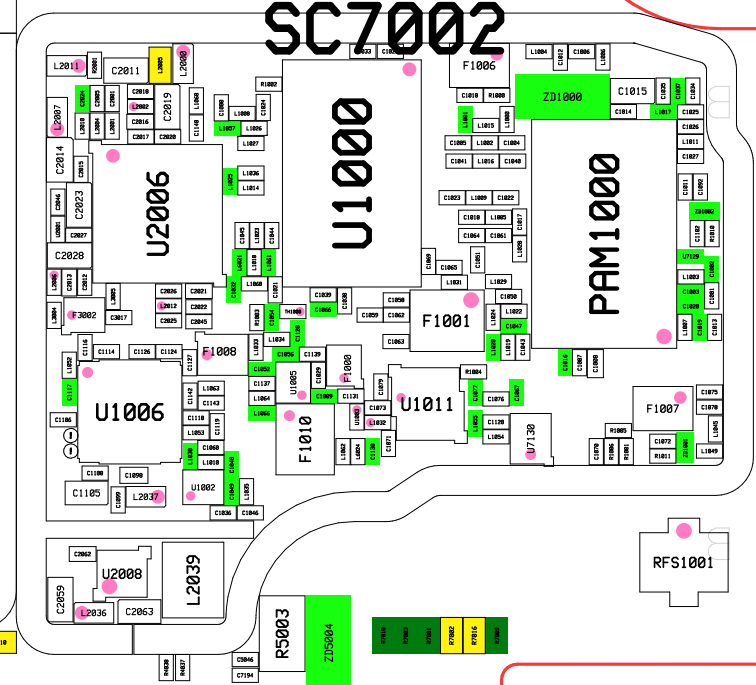
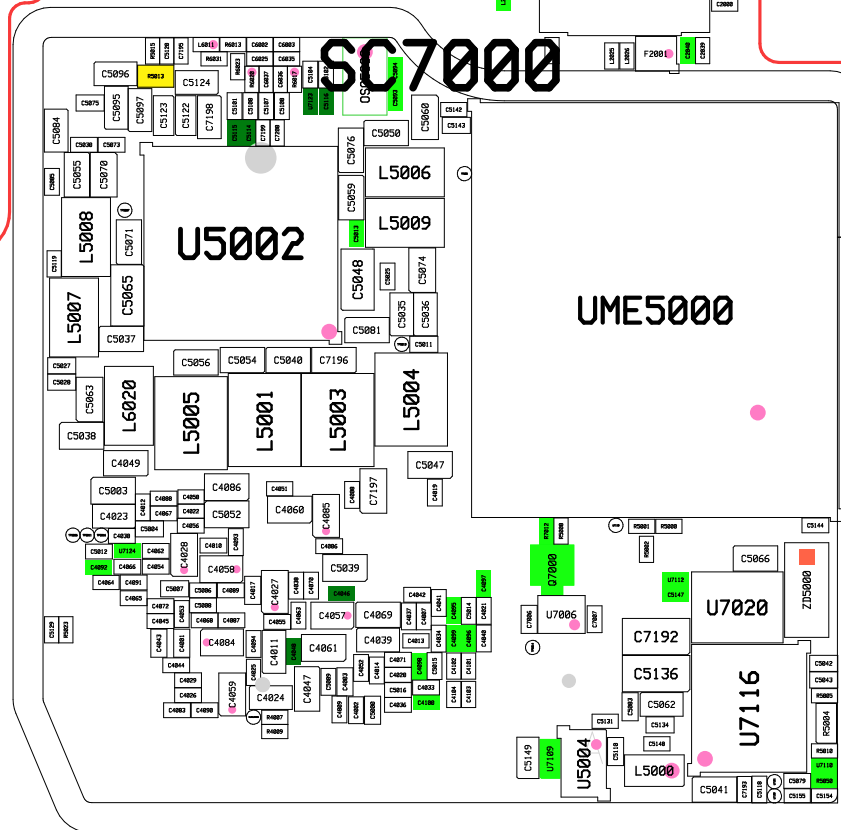
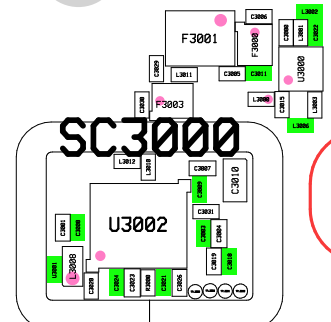
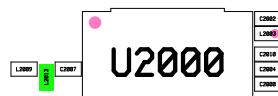
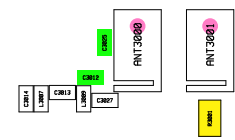
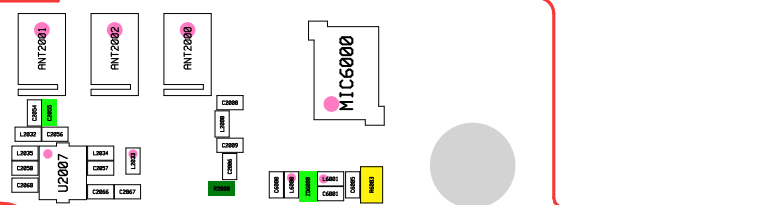
CON7000

BH7004

BH7001

JTAG7001

SM-A105G



SM-A105F REV01

2019.01.29

sheet01 : MAIN RF

sheet02 : TRANCEIVER, DRX, APT

sheet03 : GPS, BT/WIFI, FM

sheet03 : AP

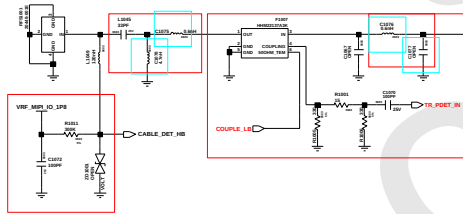
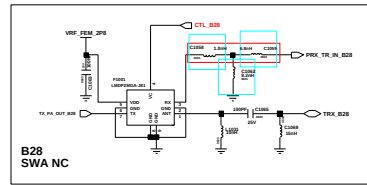
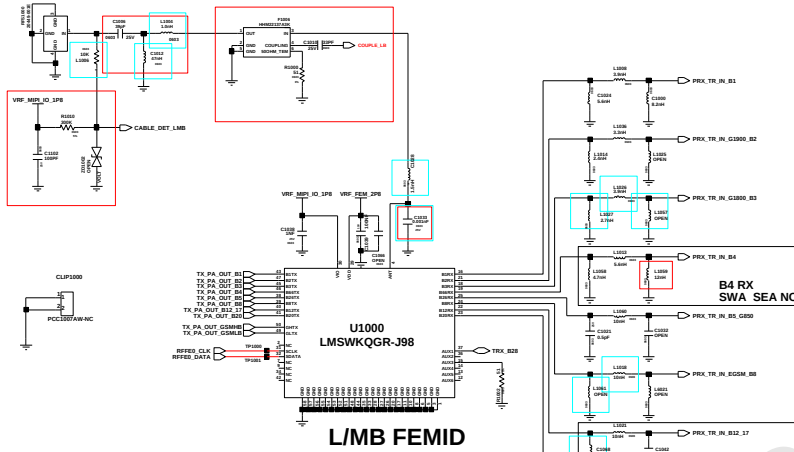
sheet05 : PMIC(Main,IF), eMMC, IF Conn, KEY, MOTOR

sheet06 : Audio, SD/SIM

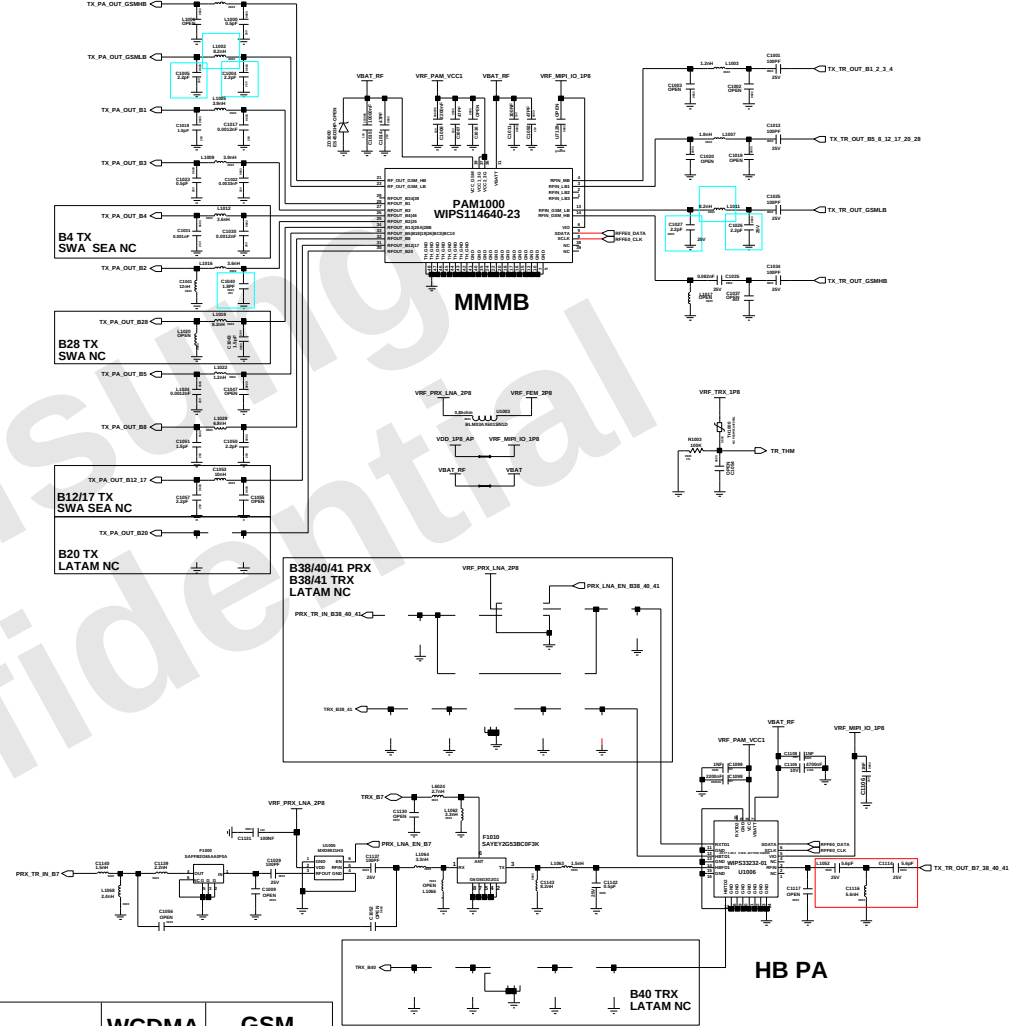
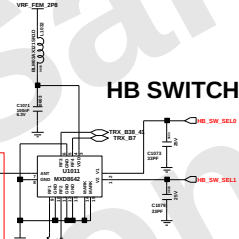
sheet07 : CAMERA, DISPLAY, SHEILD CAN,SENSOR

Author	
Checker	
Editor	
Doc. Ctrl.	
Release Ctrl.	
Version	1.0
Rev. No.	1
Rev. Date	2019.01.29

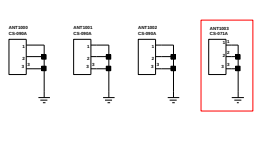
RF MAIN



HB SWITCH



GND CONTACT to BRACKET

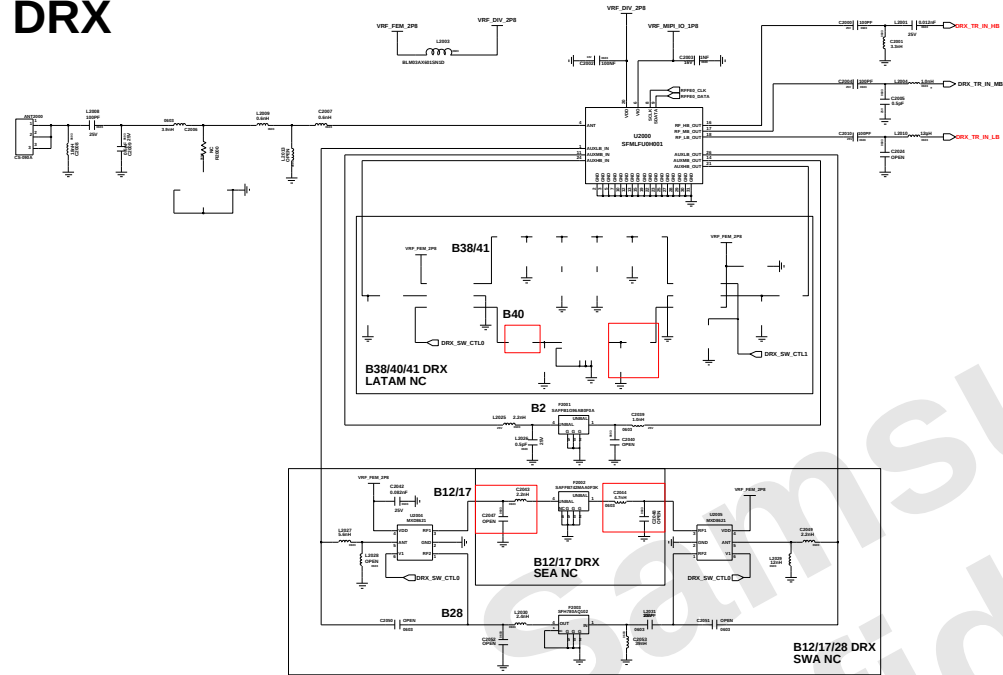


	LTE	WCDMA	GSM
SWA / CIS / MENA / AFR	B1/2/3/5/7/8/20/38/40/41	B1/2/5/8	QUAD
SEA	B1/3/5/7/8/20/28/38/40/41	B1/2/5/8	QUAD
LATAM	B1/2/3/4/5/7/8/12/17/28	B1/2/4/5/8	QUAD
KOR	B1/3/4/5/7/8/17/20/28/38/40/41	B1/2/4/5	QUAD

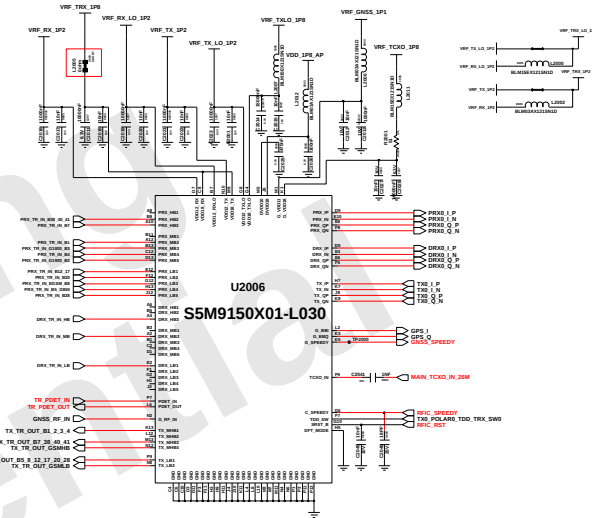
Engineer	MAIN RF
Checked by	SM-A105F
Reviewed by	
Approved by	

SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPERTY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO.'S PROPERTY
AND NOT BE LOANED TO OR REPRODUCED FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

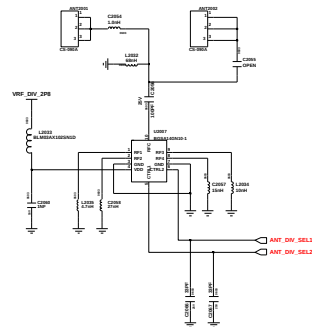
DRX



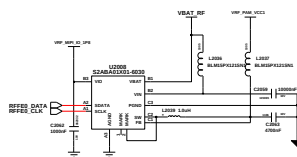
TRANSCIEVER nCA



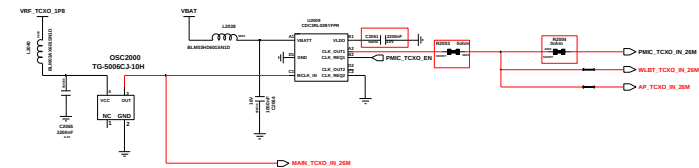
DRX GND SWITCH



APT MODULATOR

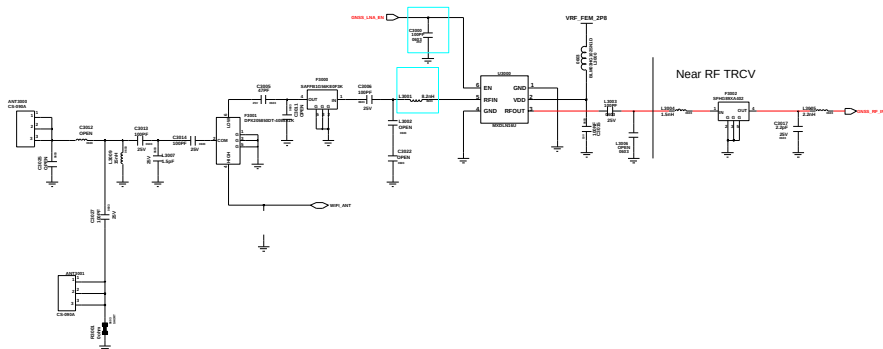


TCXO

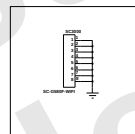
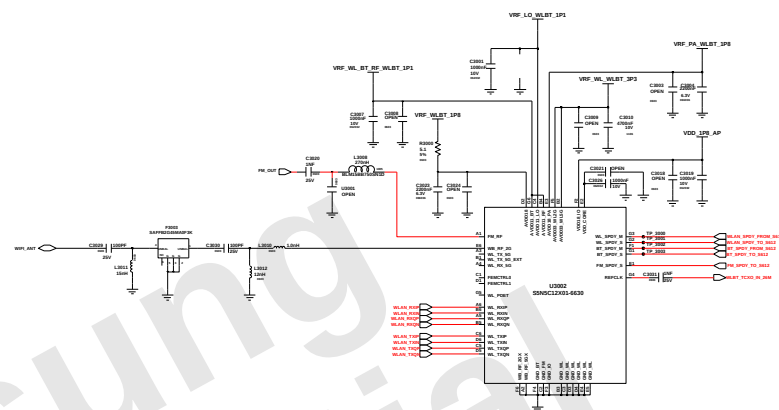


Engineer	DRX, TRANSCIEVER
Designer	
Checker	
Reviewer	
Approval	
Signature	
Date	

GNSS



BT/WIFI



Engineer:	WIFI/BT/GPS		
Drawn by:			
RED CHK:	TITLE:		
DOC & TRS CHK:	SM-A105F		
MFG ENGRG CHK:			
QA CHK:	REV	Drawing Number	Job

POP LP4 2G : SEC 1105-002944
POP LP4 3G : SEC 1105-002962

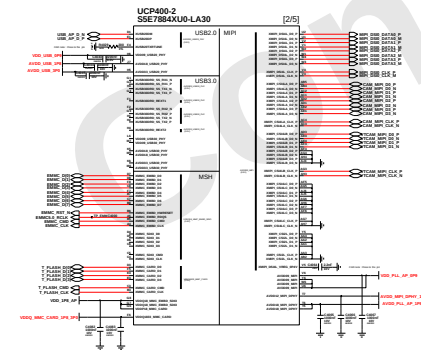
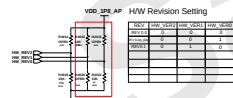
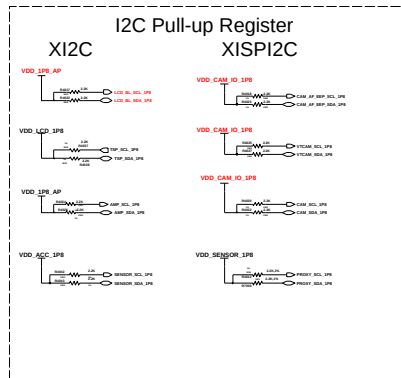
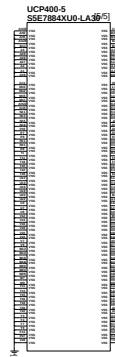
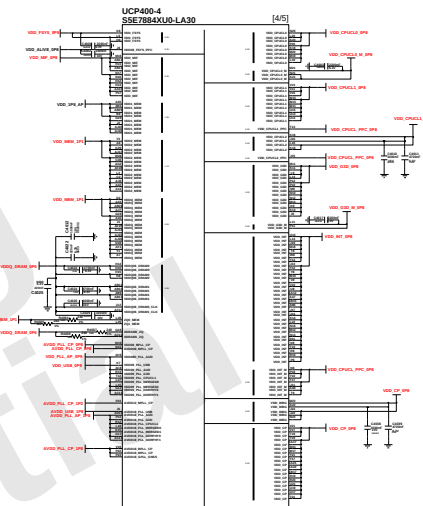
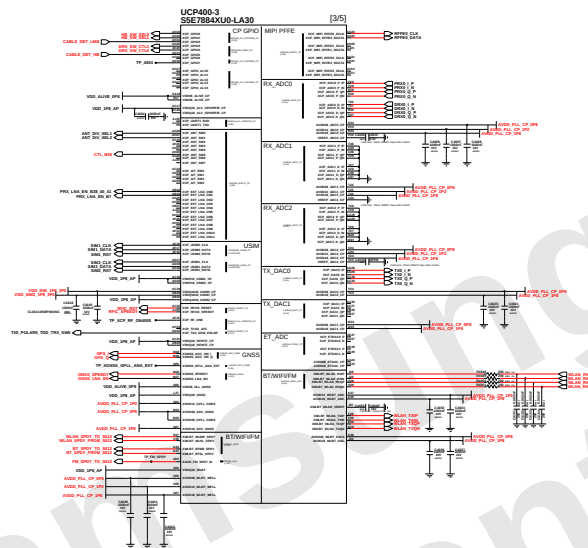
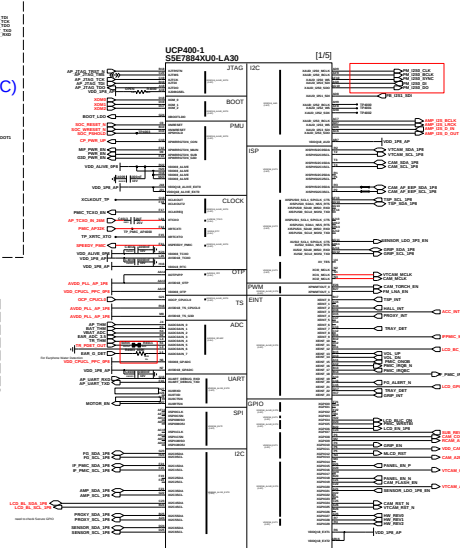
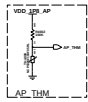
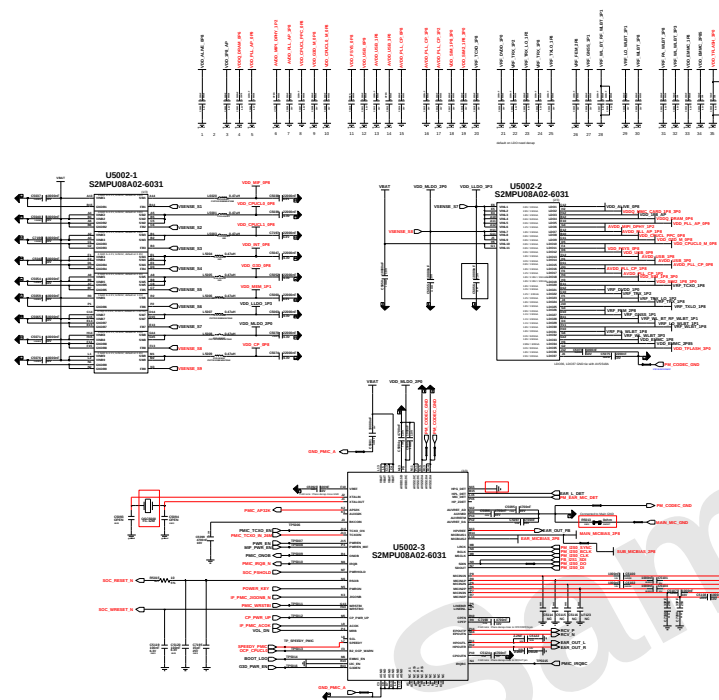


Figure 10: Recommended decoupling capacitor placement. The figure contains 12 sub-diagrams showing various decoupling capacitor connections for different power pins. Each diagram includes a VDD pin, a decoupling capacitor (C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12), and a connection to a specific ball (e.g., VDD_CPUCLOCK BALL, VDD_MF BALL, VDD_INT BALL, VDD_CPUCLOCK BALL, VDD_INT BALL, VDD_CPUCLOCK BALL, VDD_INT BALL, VDD_CPUCLOCK BALL, VDD_INT BALL, VDD_CPUCLOCK BALL, VDD_INT BALL, VDD_CPUCLOCK BALL). The diagrams illustrate the placement of capacitors relative to the power pins and the balls, with some showing multiple capacitors in parallel.

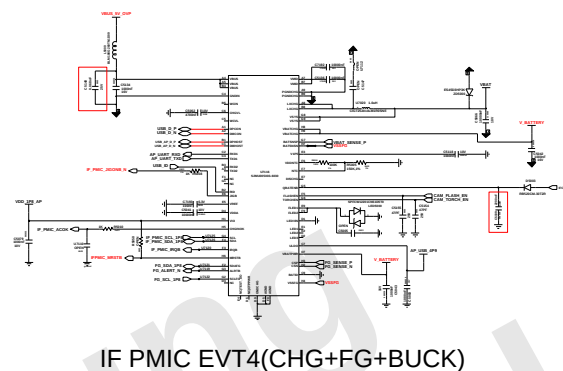
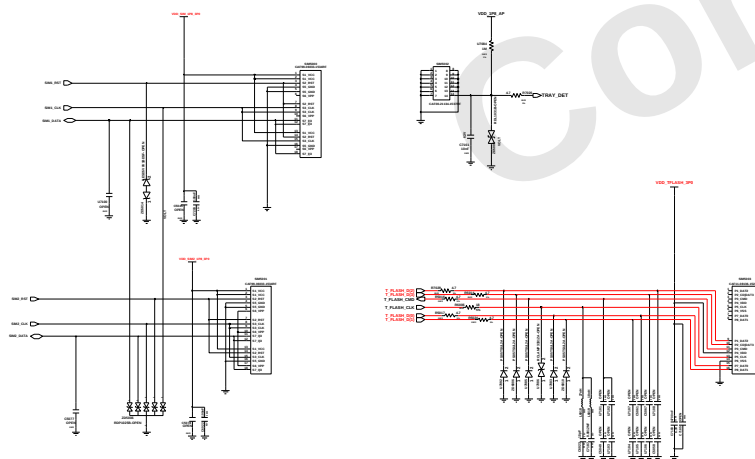
AP	
SM-A105F	

POWER

PMIC



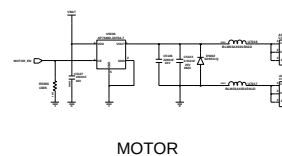
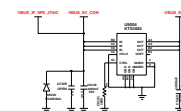
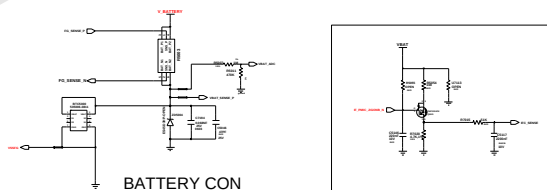
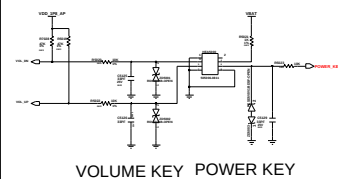
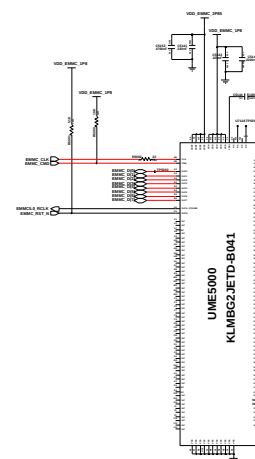
SIM & SD CARD (3IN3)



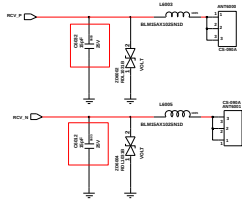
IF PMIC EVT4(CHG+FG+BUCK)



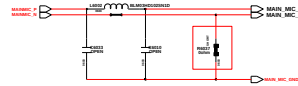
16G : SEC (1107-002548)
32G : SEC (1107-002552)



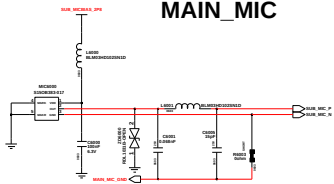
AUDIO



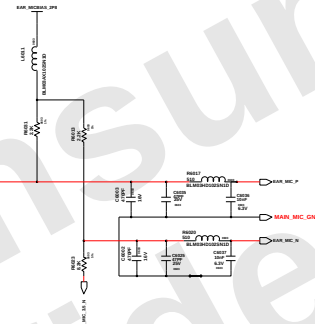
RCV Contact



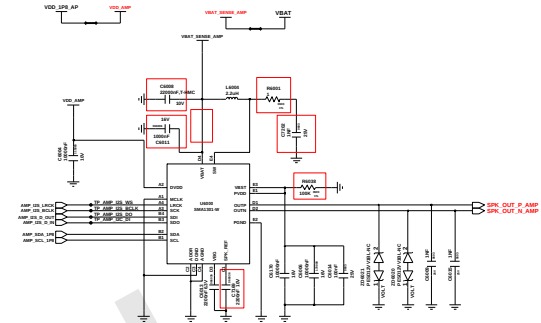
MAIN_MIC



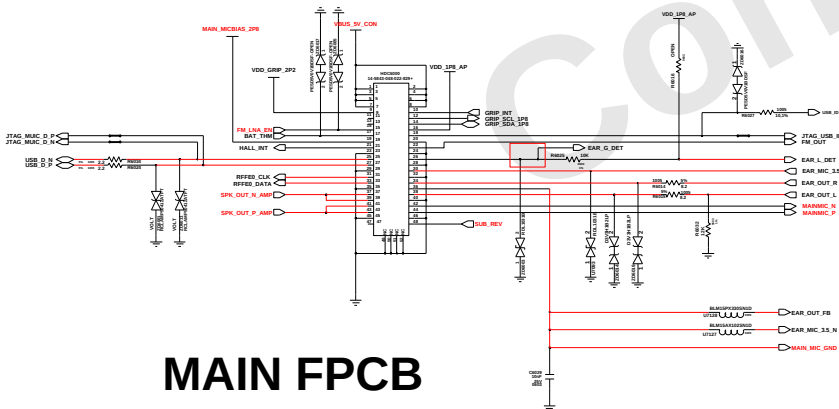
Sub MIC



EAR_MIC



Audio amp



MAIN FPCB



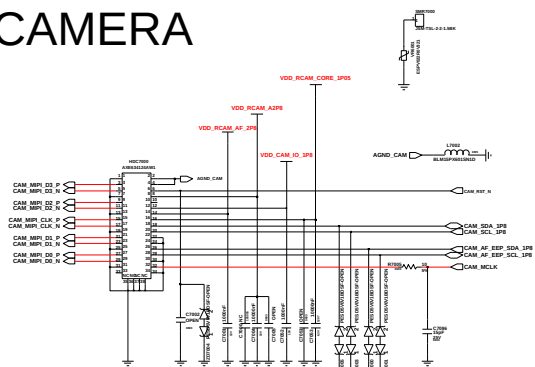
Grip Sensor_LDO

Document	AUDIO	
Version	1.0	1
File Name	SM-A105F	1
File Path	SM-A105F	1
File Size	1.0	1
File Type	PDF	1
File Date	20190222-135450	1

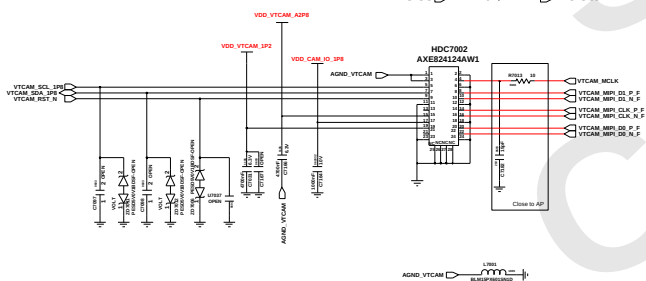
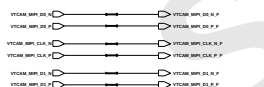
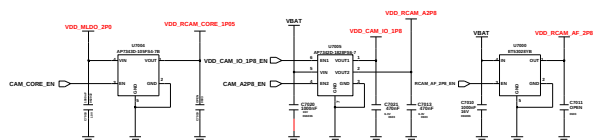
SAMSUNG CONFIDENTIAL
THIS DOCUMENT CONTAINS CONFIDENTIAL
PROPRIETARY INFORMATION THAT IS
SAMSUNG ELECTRONICS CO.'S PROPERTY
IT DISCLOSE TO OR DUPLICATE FOR OTHERS
EXCEPT AS AUTHORIZED BY SAMSUNG.

CAMERA

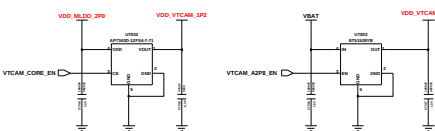
CAM DECO GASKET



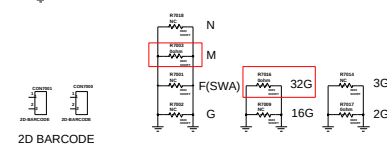
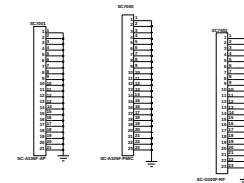
MAIN_CAMERA (13M) : SM-M205F



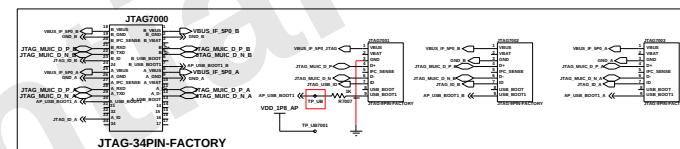
FRONT_CAMERA (5M) : SM-M105F



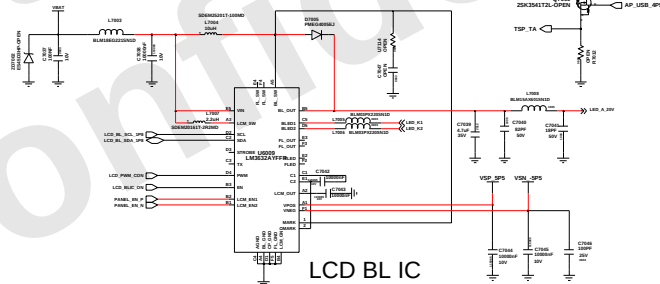
SHIELD CAN



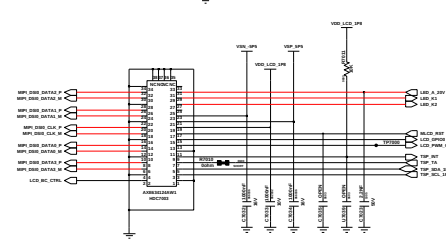
PCB JTAG



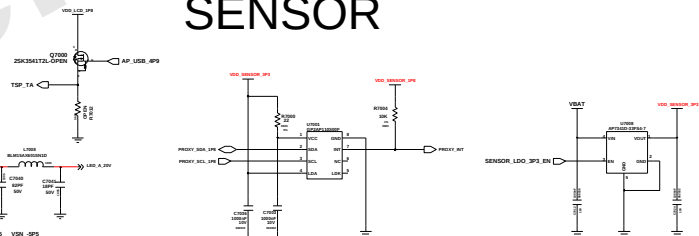
DISPLAY (6.22" INCELL LCD)



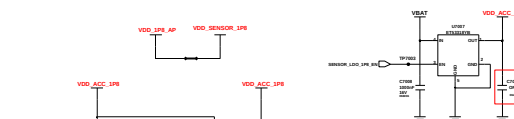
6.22" INCELL TFT LCD



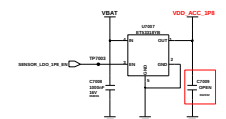
SENSOR



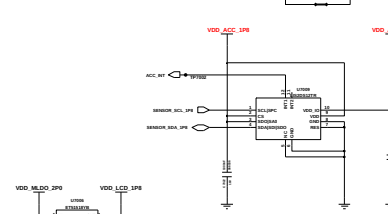
Proxy Sensor



SENSOR LDO



Accel Sensor



Engineer	CAMERA, DISPLAY, SENSOR		
Drawn by			
RED CHK	TITLE		
DOC CTRL CHK	SM-A105F		
MFG SVCR CONT			
QA CHK	REV	Drawing Number	Page

SM-A105M

U7001

BH7002

BH7005

HDC7002

SC7001

HDC7000

UCP400

SIM5002

SIM5003

SIM5000

SIM5001

HDC4000

CON7000

BH7004

BH7001

UTAG7001

